Advanced Circuits

ENGR-UH 2311 | Spring I - 2025

Experiment: Lab 2

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Lab Report: DC Operating Points Analysis of Circuits in Cadence

Introduction

The primary goal of this lab is to analyse the gain and output voltage of non-inverting and inverting amplifiers using an LM741 Op-Amp. A sine wave is used as the input signal, and the gain is determined and compared to the calculated value.

Objectives

The primary objectives of this lab includes learning to wire the circuits involving +Vcc and -Vcc on a breadboard, applying the concept of a ground in an electrical circuit, troubleshooting and analysing the circuits, operating different desktop instruments such as Multimeter, Function Generators, Oscilloscopes etc, and reading the datasheets of different component/ICs to identify the characteristics and the pinouts of the component/ICs.

Materials and Equipment

1. LM741 Operational Amplifier
2. Resistors: 91 kΩ and 2.4 kΩ
3. DC Power Supply
4. Oscilloscope
5. Function Generator
6. Hand-held Multi-meter
7. Breadboard
8. Jumper Wires

Diagram

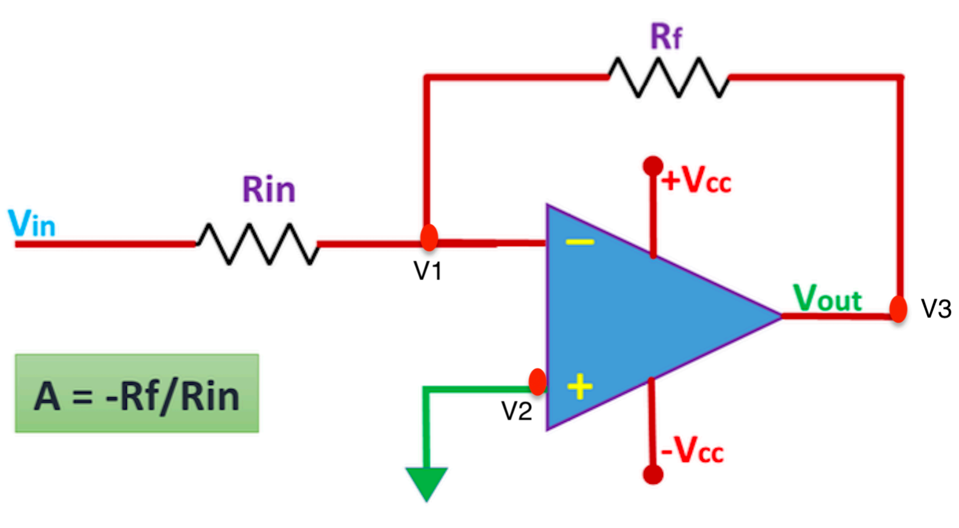


Figure 1: Schematic Diagram for Inverting Amplifier

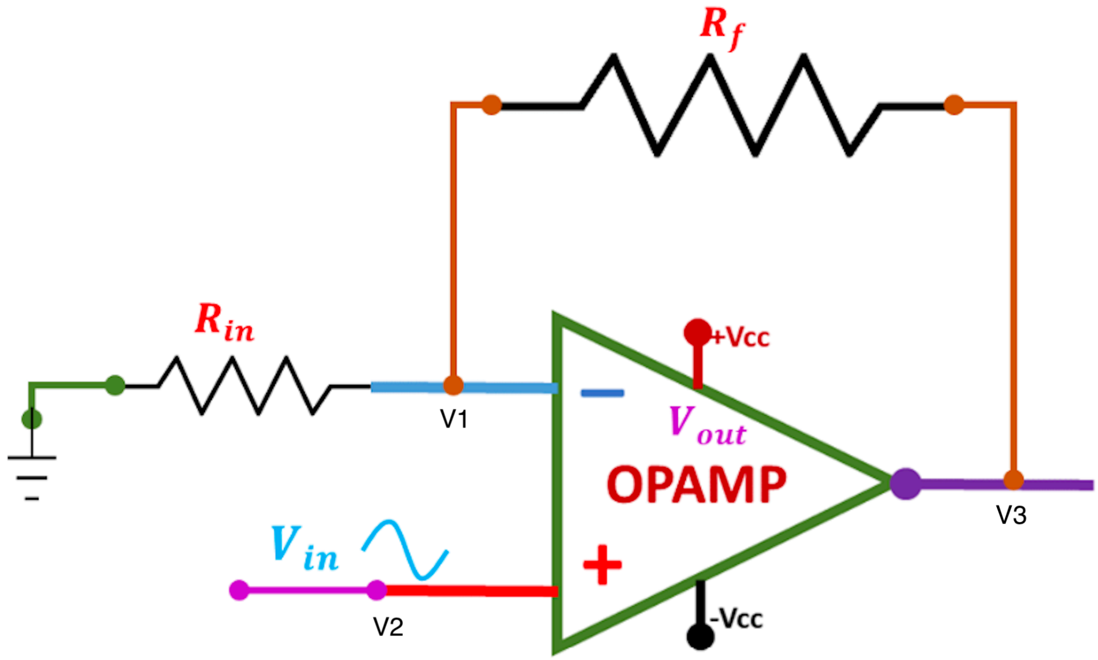


Figure 2: Schematic Diagram for Non-Inverting Amplifier

Procedure

1. Using the datasheet, the pinouts of the IC of the LM741 Operational Amplifier were identified.

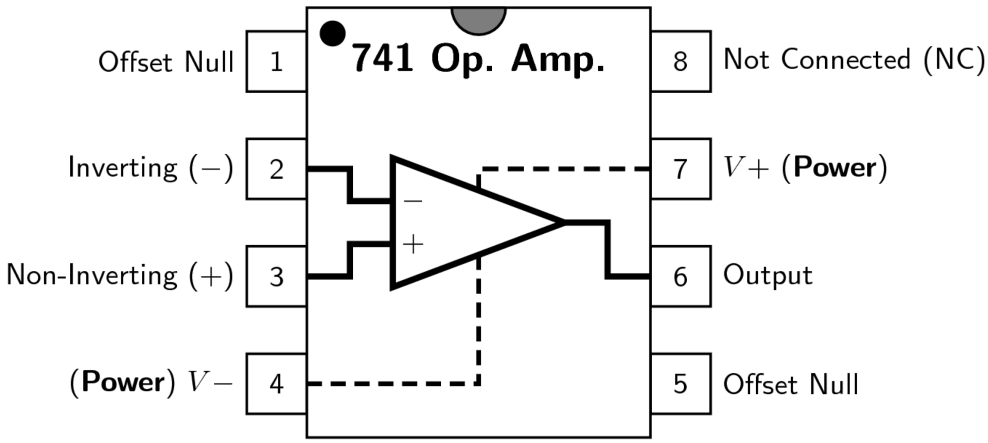


Figure 3: Pinout Diagram of LM741 Op-Amp from Datasheet.

2. An Inverting Amplifier was constructed using a breadboard following the schematic above where Rf = 91 kΩ and Rin = 2.4 kΩ.

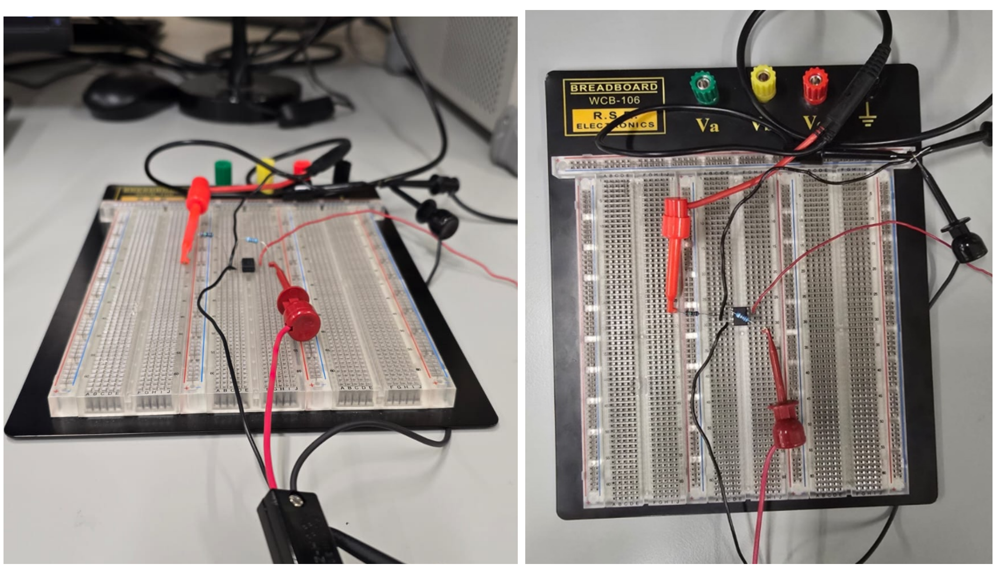


Figure 4: Breadboard set up of Inverting Amplifier Circuit

3. The input signal was then connected to the inverting (-) input of the Op-Amp while the non-inverting (+) input was grounded.

4. The power was set : $+V_{cc} = +8$ V and $-V_{cc} = -8$ V; and the current of the Power supply was limited to 100 mA ONLY for both using the DC supply.

5. A function generator was then used to generate a sinusoidal wave with a peak-to-peak amplitude of 100 $mV_{p\text{-}p}$ and a frequency of 1kHz.

6. Using a BNC-BNC cable to connect the Function Generator to Channel 1 of the Oscilloscope and another BNC-BNC cable to connect the output of the amplifier to Channel 2 of the Oscilloscope.

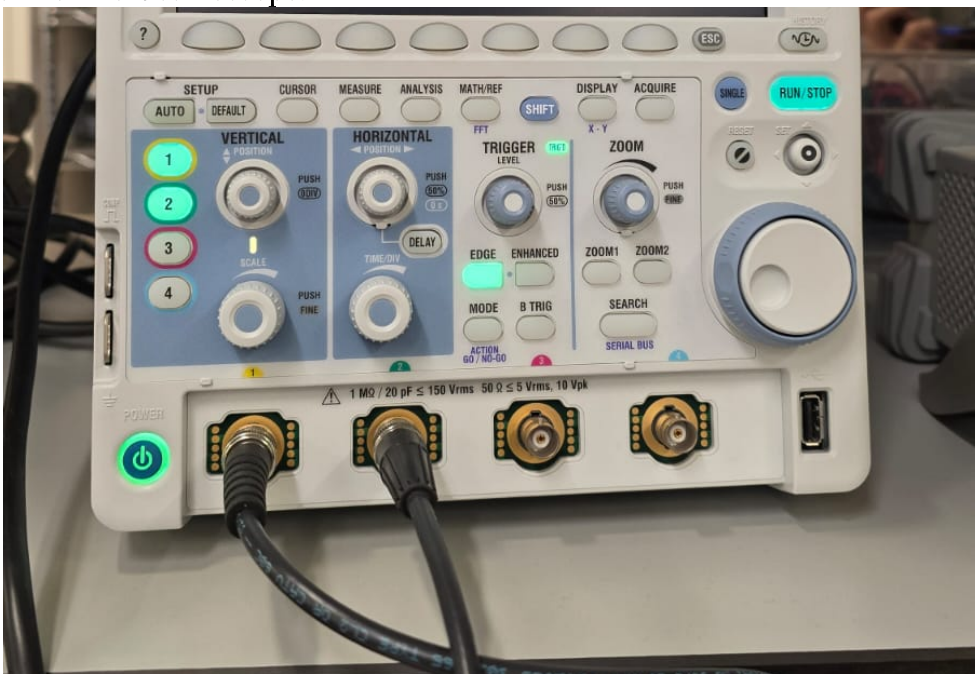


Figure 5: Input and Output Channel Set Up Using BNC Cable

7. The input and output signals on the Oscilloscope was observed.
8. The working of the Inverting Amplifier was verified by calculating the gain of the amplifier using the formula:

$$A_v = -\frac{R_f}{R_{in}}$$

9. The procedure was repeated for the Non-Inverting Amplifier following the schematic above.
10. The input signal was then connected to the non-inverting (+) input of the Op-Amp while the inverting (-) input was grounded.

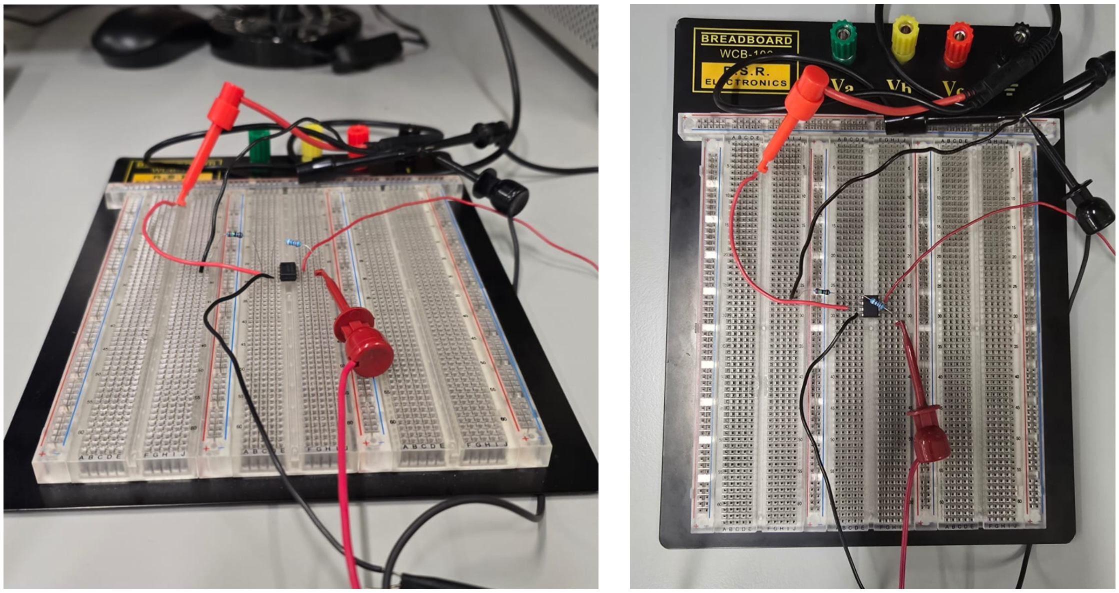


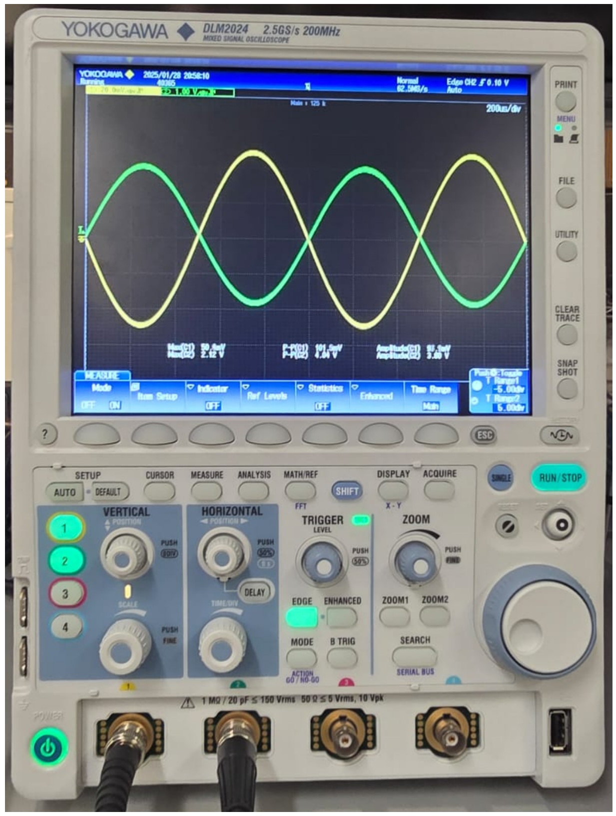
Figure 6: Breadboard set up of Non-Inverting Amplifier Circuit

11. - The gain of the non-inverting amplifier was then calculated using the formula:

$$A_v = 1 + \frac{R_f}{R_{in}}$$

Results

INVERTING AMPLIFIER



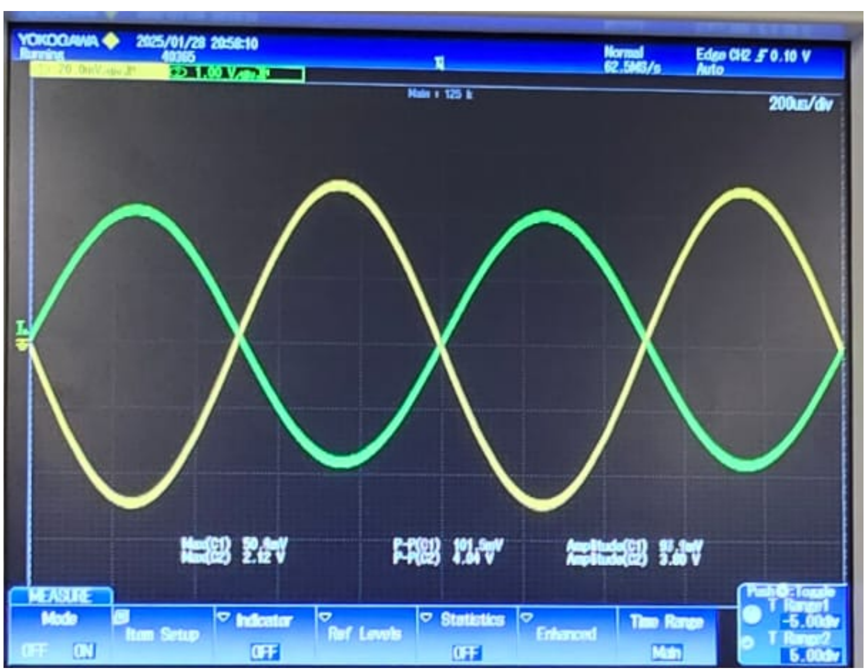


Figure 7: Output and Input Waveform Plots on Oscilloscope for Inverting Amplifier

Calculated Gain of Inverting Amplifier

$A_v = -\frac{R_f}{R_{in}} = -\frac{91\text{k}\Omega}{2.4\ \text{k}\Omega} = -37.92$

Gain of Inverting Amplifier Obtained from Oscilloscope

Gain, $A = -\frac{\text{Vout}}{\text{Vin}} = -\frac{4.01V}{101.5\ \text{mV}} = -39.51$

NON INVERTING AMPLIFIER

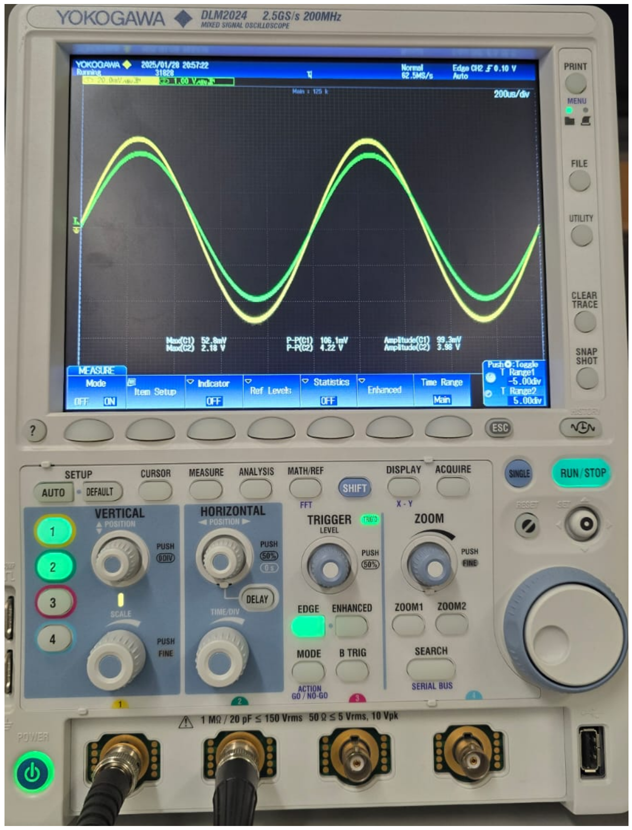


Figure 8: Output and Input Waveform Plots on Oscilloscope for Non-Inverting Amplifier

Calculated Gain of Non-Inverting Amplifier

$A_v = 1 + \frac{R_f}{R_{in}} = 1 + \frac{91\text{k}\Omega}{2.4\ \text{k}\Omega} = 38.92$

Gain of Non-Inverting Amplifier Obtained from Oscilloscope

Gain, $A = \frac{\text{Vout}}{\text{Vin}} = \frac{4.22V}{106.3\ \text{mV}} = 39.70$

Discussion

The gains obtained from the experiment were close to the theoretical values. The calculated gain of the inverting amplifier was -37.92, while the gain obtained from the oscilloscope was -39.51. Similarly, the calculated gain of the non-inverting amplifier was 38.92, while the gain obtained from the oscilloscope was 39.70. These results show that the measured gains closely matched the theoretical values for lower input amplitudes.

However, the experiment also demonstrated the importance of considering power supply limitations when designing and operating op-amp circuits. *Saturation* in Op-Amps occurs when the output voltage of the circuit exceeds the possible range. In this case, the op-amp outputs its maximum or minimum possible voltage instead, resulting in a clipped signal. This is because the op-amp is limited by its power supply voltages, which are often referred to as the "rails". When the op-amp output saturation causes the signal to be cut off close to the rails, we say that the signal is clipped.

To illustrate the effect of saturation, let's consider an example where the input voltage is increased to 200 mV_{p-p}. For the inverting amplifier, the output voltage would be:

$$V_{out} = -39.51 \times 200\ mV_{p-p} = -7.9\ V_{p-p}.$$

Since the power supply limits are +/- 8V, the output voltage would be clipped close to 8V, resulting in a distorted waveform as shown in the figure below.

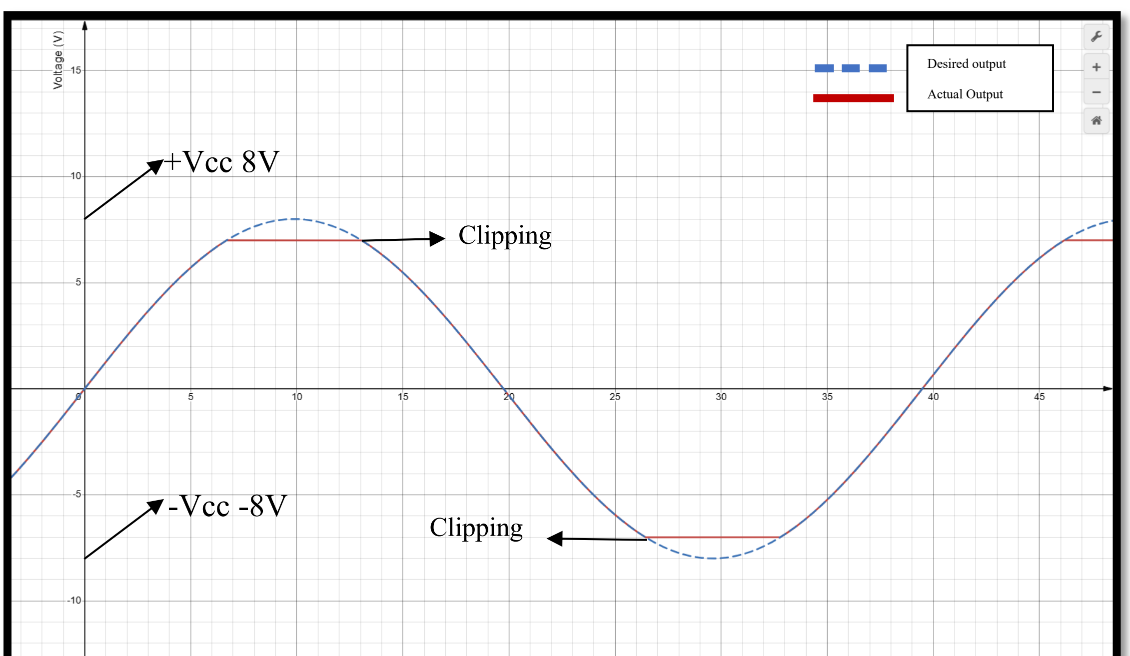


Figure 9: Saturated Output Voltage Waveform with 200mV for Inverting Amplifier

For the non-inverting amplifier, the output voltage would be:

$$\text{Vout} = 39.70 \text{ x } 200 \text{ mV}_{p\text{-}p} = 7.94 \text{ V}_{p\text{-}p}.$$

Again, since the power supply limits are +/- 8V, the output voltage would be clipped close to 8V, resulting in a distorted waveform as shown in the figure below.

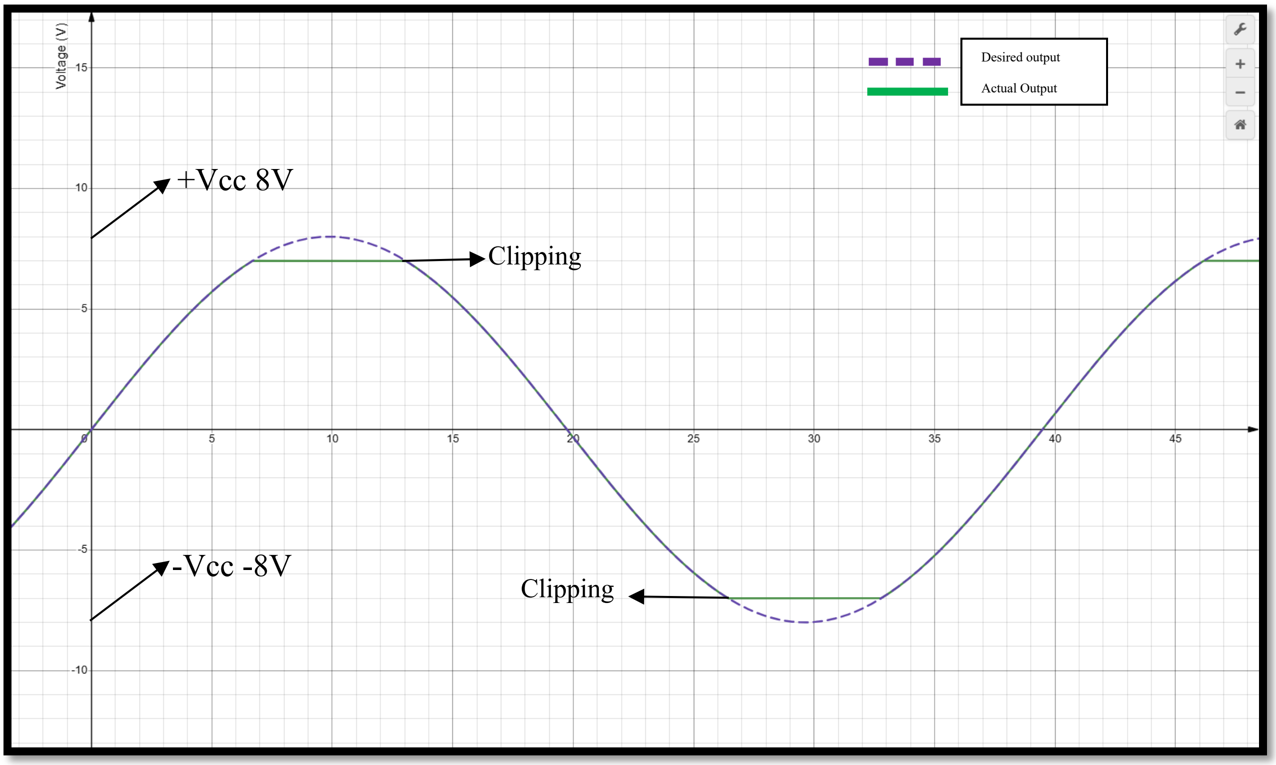


Figure 10: Saturated Output Voltage Waveform with 200 mV input Voltage for Nom-Inverting Amplifier

It is important to note that for ideal operational amplifiers, the clipping takes place exactly at 8V, so in the case of the 200 mV input signal and the gains obtained, a normal output wave would be expected. However, this is not so for the real case. Real op-amps consists of components that must be powered for the op-amp to function. As a result there is voltage drop across these components which result in the clipping voltage being within the rails and not exactly at the supply as it is for the ideal case.

The output waveforms for both amplifiers would be clipped close to their supply voltages of +/- 8V . This is because the op-amp is limited by its power supply voltages, and the output voltage cannot exceed these limits. The reason for the observed output waveform is due to the saturation of the op-amp. When the input voltage is increased, the output voltage also increases, but it is limited by the power supply voltages. As a result, the output waveform becomes clipped, resulting in a distorted signal.

Conclusion

The experiment demonstrated the importance of considering power supply limitations when designing and operating op-amp circuits. The results showed that the measured gains closely matched the theoretical values for lower input amplitudes, but distortion occurred at higher amplitudes due to saturation of the op-amp output. The output waveforms for both amplifiers were clipped when the input voltage was increased, resulting in distorted signals.